Key vocabulary

Properties: The way a substance looks and behaves.

States of matter: Solids, liquids and gases.

Density: The mass of something compared to its volume.

Key information

The three states of matter; **solids**, **liquids** and **gases**, all have different properties. This is because of the way the particles are arranged inside the substances.

Task One

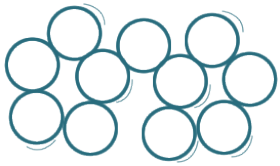
Read the description of how the particles are arranged in each state of matter and draw the particle model in the box.

Solid particle arrangement The particles are very close together in an ordered arrangement. The particles vibrate in a fixed position. The particles have very strong forces between them.	
Liquid particle arrangement The particles are close together. The particles are free to move below the surface. The particles vibrate and move round each other. The particles have quite strong forces between them.	
Gas particle arrangement The particles are far apart in a random arrangement. The particles are free to move anywhere. The forces between the particles are weak.	

Task two

Cut out the boxes and use then to complete the table.

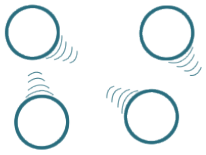
	Solids	Liquids	Gases
Particle model			
Particle arrangement			.
Forces between particles			



Very strong forces between bonds.

Particles are very close together in an ordered arrangement.

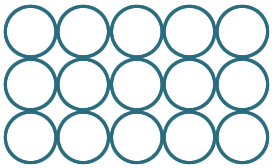
Quite strong forces between bonds.



Particles are far apart in a random arrangement.

Weak forces between bonds.

Particles are quite close together but can move round each other.



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Task three

Match the properties to the correct state of matter and write them in the correct box. Remember that states of matter can have the same properties.

no fixed shape

fixed volume

very dense

no fixed volume

fixed shape

takes the shape of the container

low density

fixed volume

quite dense

State of matter	Properties
Solids	
Liquids	
Gases	

Teaching notes and answers

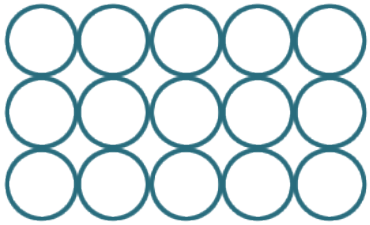
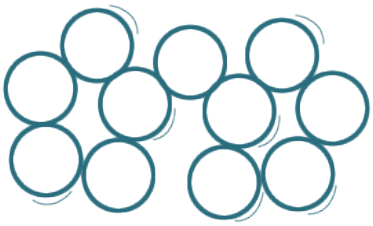
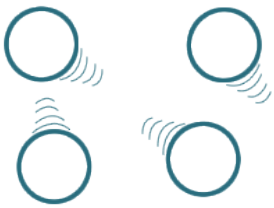
This activity sheet introduces some of the key vocabulary and knowledge required for students to understand **particle theory**.

Task one is a short comprehension exercise. Students must use and interpret the information provided to draw the particle model.

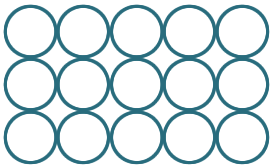
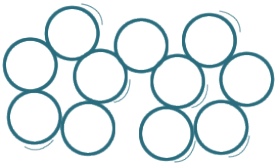
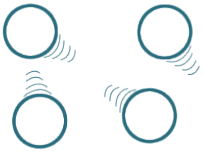
Task two is to check the students' knowledge and understanding of task one and to further cement their learning.

Task three links the particle model of the states of matter to the properties of that state.

Task one – answers

Solid particle arrangement The particles are very close together in an ordered arrangement. The particles vibrate in a fixed position. The particles have very strong forces between them.	
Liquid particle arrangement The particles are close together. The particles are free to move below the surface. The particles vibrate and move round each other. The particles have quite strong forces between them.	
Gas particle arrangement The particles are far apart in a random arrangement. The particles are free to move anywhere. The forces between the particles are weak.	

Task two – answers

	Solids	Liquids	Gases
Particle model			
Particle arrangement	Particles are very close together in an ordered arrangement.	Particles are quite close together but can move round each other.	Particles are far apart in a random arrangement.
Forces between particles	Very strong forces between bonds.	Quite strong forces between bonds.	Weak forces between bonds.

Task three – answers

State of matter	Properties
Solids	fixed shape very dense fixed volume
Liquids	takes the shape of the container quite dense fixed volume
Gases	no fixed shape low density no fixed volume